

AI SOFTWARE DEVELOPMENT INTERNSHIP

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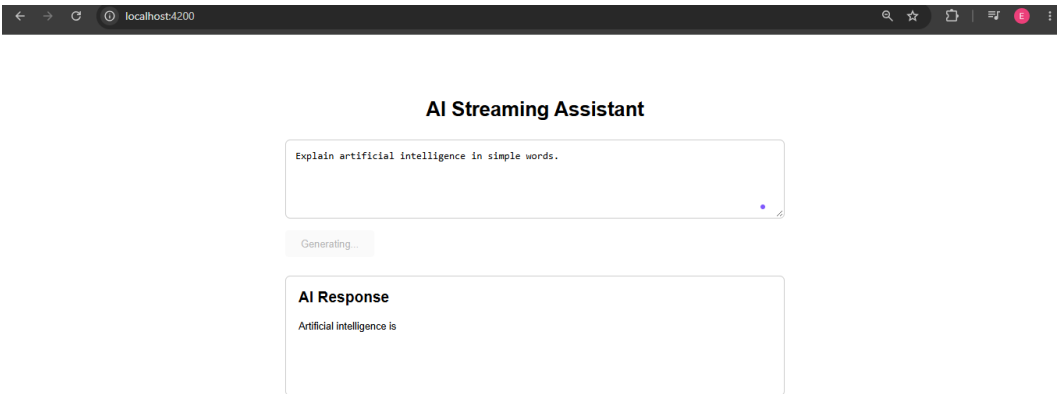
Week 6 — Detailed Task Sheet

LangChain & LCEL • Advanced Retrieval • Tool Calling Resilient .NET fi AI
Service Integration • End-to-End Streaming • Git Rebase

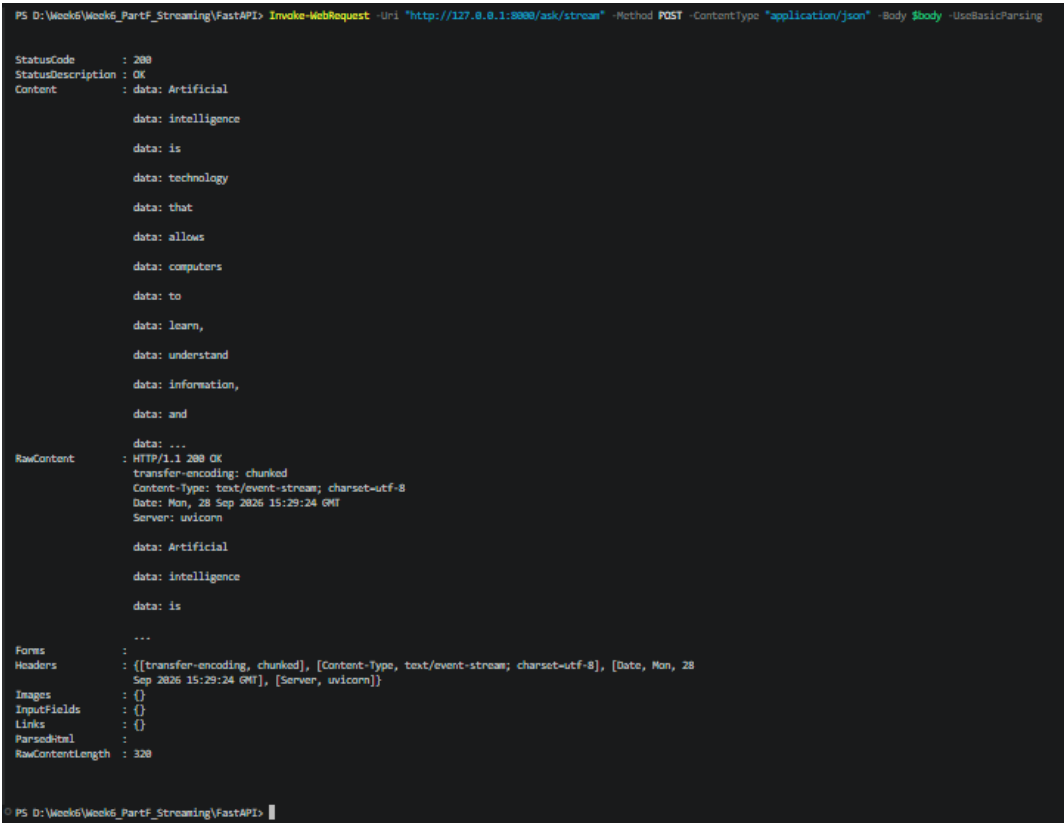
PART F — END-TO-END STREAMING (FASTAPI fi .NET fi ANGULAR)

Outputs

Visible Token-by-Token Rendering in Angular



.NET Proxy Streaming FastAPI Response



```

PS D:\Weeks\Week6_PartF_Streaming\DotNetProxy> $body = @( question - "Explain artificial intelligence in simple words." ) | ConvertTo-Json
PS D:\Weeks\Week6_PartF_Streaming\DotNetProxy> Invoke-WebRequest -Uri "http://localhost:5173/api/assistant/ask/stream" -Method POST -ContentType "application/json" -Body $body -UseBasicParsing

StatusCode      : 200
StatusDescription : OK
Content         : data: Artificial
                  data: intelligence
                  data: is
                  data: technology
                  data: that
                  data: allows
                  data: computers
                  data: to
                  data: learn,
                  data: understand
                  data: information,
                  data: and
                  data: ...
RawContent      : HTTP/1.1 200 OK
                  Transfer-Encoding: chunked
                  Cache-Control: no-cache
                  Content-Type: text/event-stream
                  Date: Mon, 28 Sep 2026 15:42:18 GMT
                  Server: Kestrel
                  data: Artificial
                  data: intelligence
                  d...
Forms           : 
Headers         : {[Transfer-Encoding, chunked], [Cache-Control, no-cache], [Content-Type, text/event-stream], [Date, Mon, 28 Sep 2026 15:42:18 GMT]...}
Images          : 
InputFields     : 
Links           : 
ParsedHtml      : 
RawContentLength : 320

```

FastAPI Deliberate 0.3s Streaming Delay

```

@app.post("/ask/stream")
def ask_stream(request: AskRequest):
    def event_generator():
        # Temporary test response
        # This proves real SSE token-by-token streaming.
        words = (
            "Artificial intelligence is technology that allows computers "
            "to learn, understand information, and perform tasks "
            "that normally require human intelligence."
        ).split()

        for word in words:
            time.sleep(0.3)
            yield f"data: {word} \n\n"

        yield "data: [DONE]\n\n"

    return StreamingResponse(
        event_generator(),
        media_type="text/event-stream"
    )

```



AI Streaming Assistant

Explain artificial intelligence in simple words.

Ask AI

AI Response

Artificial intelligence is technology that allows computers to learn, understand information, and perform tasks that normally require human intelligence.

Challenge 3: FlushAsync() removed — buffering behavior observed



AI Streaming Assistant

Explain artificial intelligence in simple words.

Generating...

AI Response

Artificial intelligence is



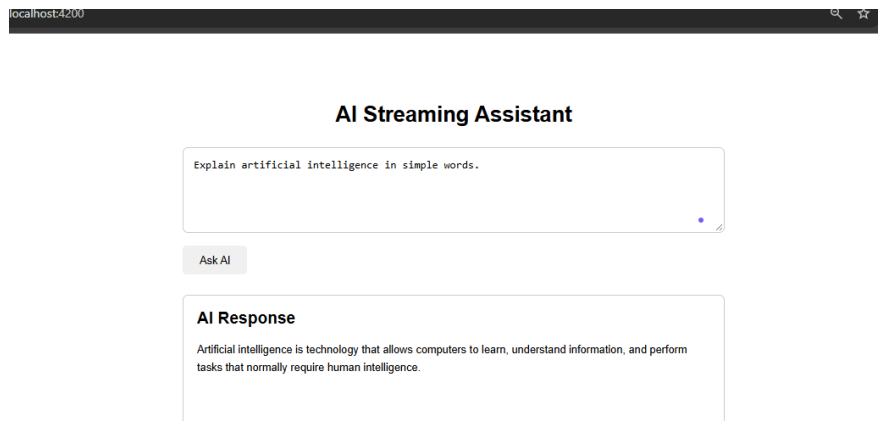
AI Streaming Assistant

Explain artificial intelligence in simple words.

Generating...

AI Response

Artificial intelligence is technology that allows computers

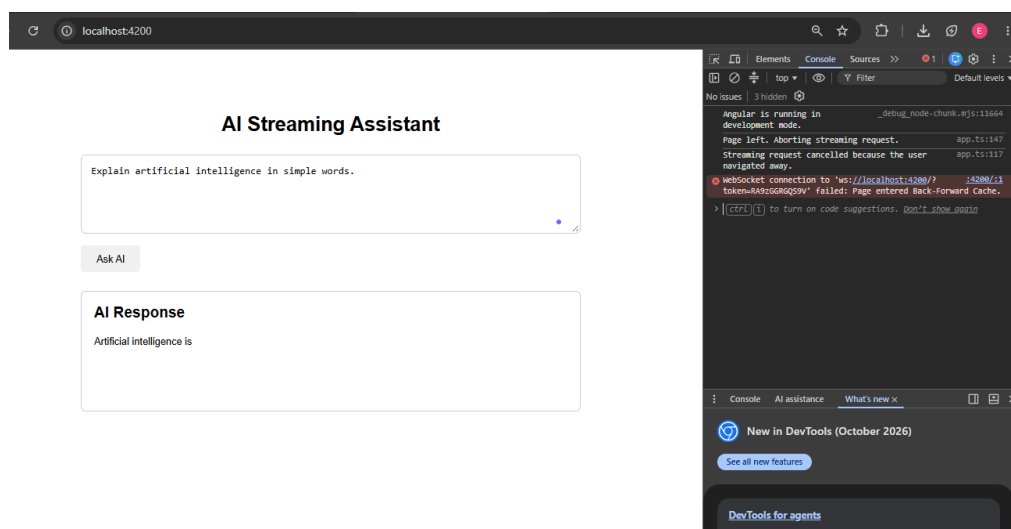


FlushAsync() Experiment:

I temporarily removed `FlushAsync()` from the .NET streaming proxy and tested the application. The response no longer had the same immediate flushing behavior and appeared in delayed chunks rather than being pushed as promptly to the browser. This demonstrated that writing data to the response does not necessarily mean it is immediately sent to the client.

I then restored `FlushAsync()` after each `WriteAsync()` call. `FlushAsync()` explicitly asks the response body to flush the currently written data, which helps the streamed chunks reach Angular promptly. Therefore, it is important for achieving responsive real-time streaming behavior in the .NET proxy.

Navigation_Away_Cancellation



```
Problems Output Debug Console Terminal Ports
PS D:\Week6\Week6_PartF_Streaming\DotNetProxy> dotnet run
om\dotnet\rundamentals\code-analysis\quality-rules/ca2924)
info: Microsoft.Hosting.Lifetime[14]
    Now listening on: http://localhost:5173
info: Microsoft.Hosting.Lifetime[0]
    Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
    Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
    Content root path: D:\Week6\Week6_PartF_Streaming\DotNetProxy
info: System.Net.Http.HttpClient.AIService.LogicalHandler[100]
    Start processing HTTP request POST http://127.0.0.1:8000/ask/stream
info: System.Net.Http.HttpClient.AIService.ClientHandler[100]
    Sending HTTP request POST http://127.0.0.1:8000/ask/stream
info: System.Net.Http.HttpClient.AIService.ClientHandler[101]
    Received HTTP response headers after 143.1087ms - 200
info: System.Net.Http.HttpClient.AIService.LogicalHandler[101]
    End processing HTTP request after 176.9033ms - 200
CANCELLATION TEST: Client disconnected. Upstream AI streaming request was cancelled.
```